

Intro to AI for Medicine

Lec 6: Network

Last Time:

- Discussed different ways of encoding data so it can interface with matrices
- Went over types of bias
- Introduced data augmentation

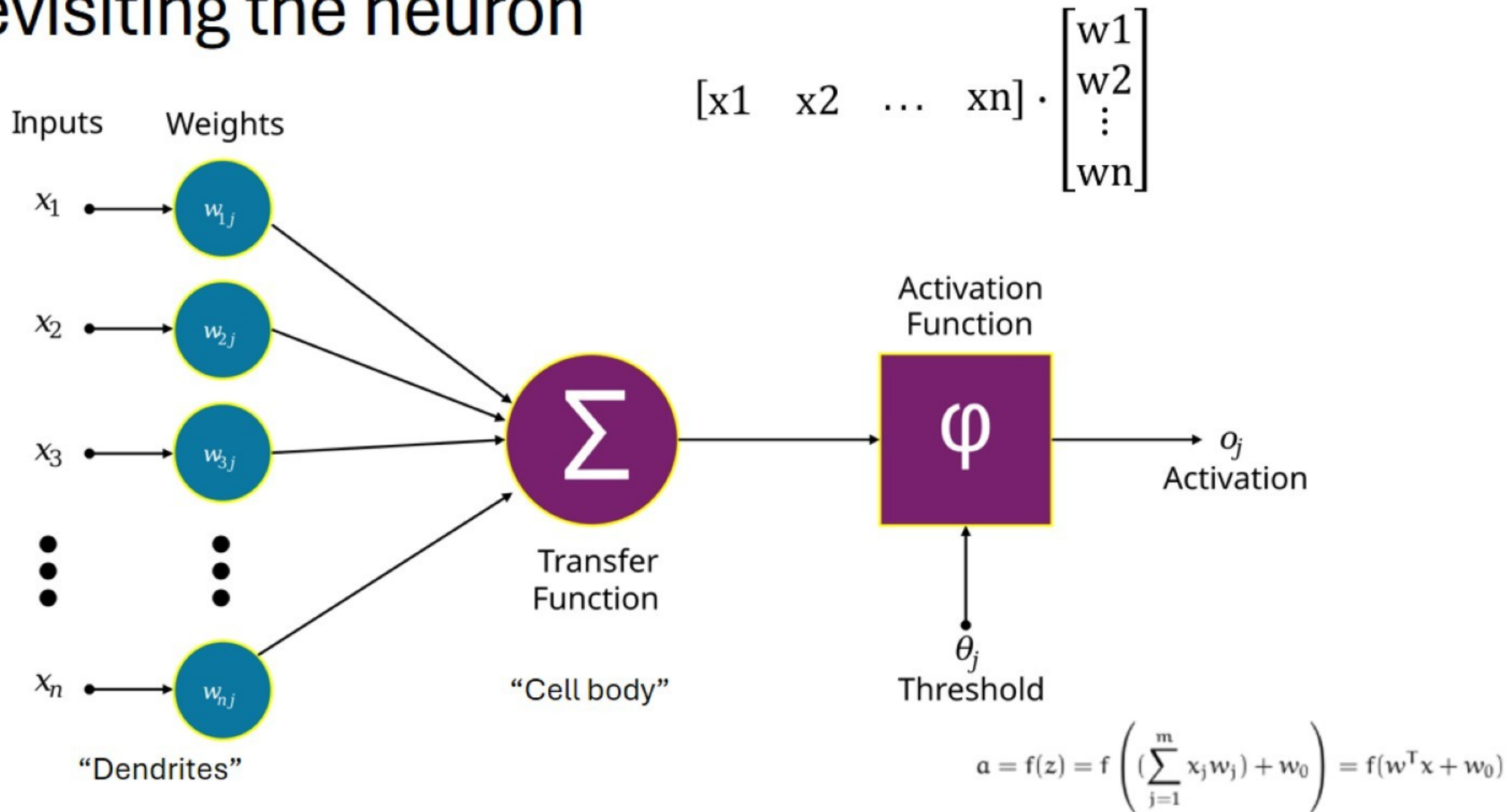
Before

- Learned what the structure of the regular neural net looks like

Today

- Variations on the theme of the regular neural net, and how that can help solve a wider class of problems
- Everything we'll see today is just a small but creative tweak to the network we already know!
- Each tweak addresses a key limitation with the original NN for fitting different problems

Revisiting the neuron

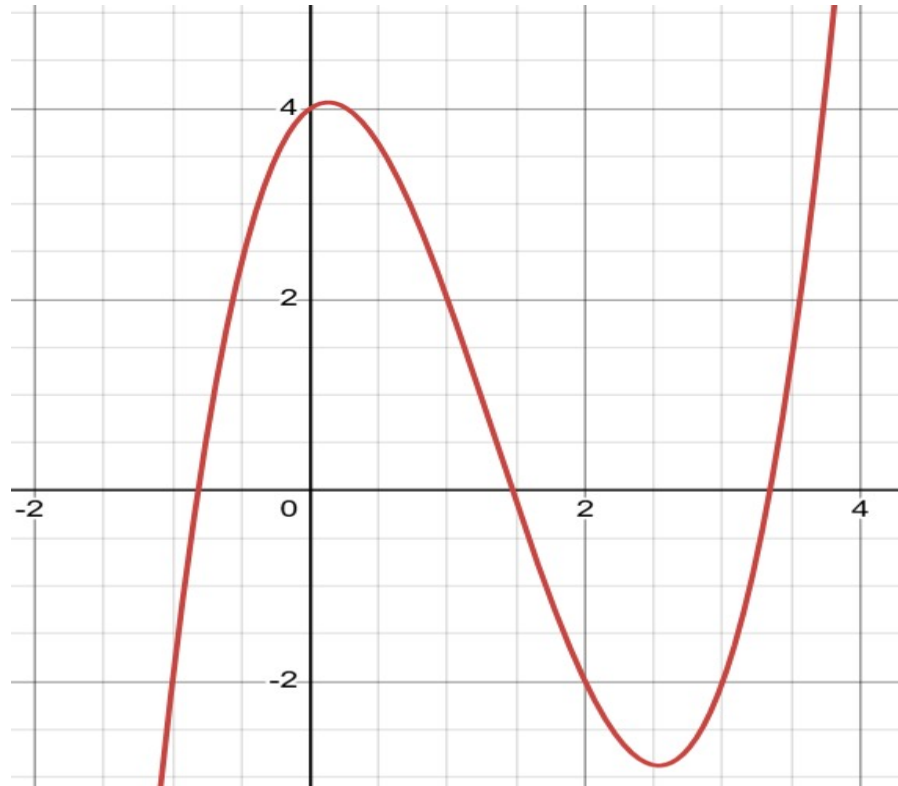


Inductive Bias

- Not bias in the traditional sense – it's the direction we point our model toward via the construction of the network
- Ex. For linear regression (no neural net) we assume our data can be fit linearly
- For other problems, we can give our ML model a hint by changing the way it processes data

Examples of Inductive Bias

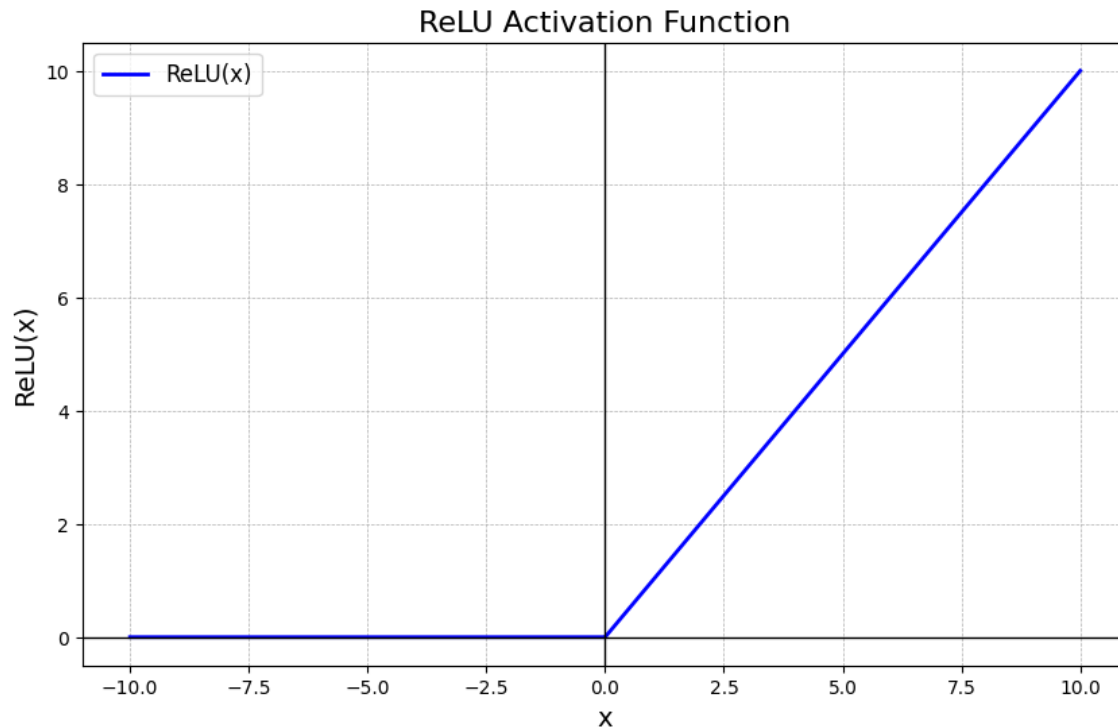
- Smoothness: Similar inputs have similar outputs



Examples of Inductive Bias

- Sparsity: Your question can be answered by only a few key inputs

• In other words: not every input matters for every question



$$+ \underbrace{\lambda \cdot \sum_{j=1}^p |b_j|}_{\text{Regularizer term}}$$

Inductive Bias

- No free lunch rule: If your inductive bias works well for one type of data, it works worse for another type.

Can We:

- Incorporate time?
- Extract only the most key information, removing noise?
- Our outputs want to be as similar to a known dataset as possible?
- Consider spatial relationships in imaging?
- Recurrent Neural Nets
- Autoencoders
- Generative Adversarial Nets
- Convolutional Nets

Example Case

- Chart data for patients in the PICU
- Every day, we get a new set of labs
- Every few hours, we get new vitals
- Can we use not only the data, but the change in the data to accurately diagnose?

LEARNING TO DIAGNOSE WITH LSTM RECURRENT NEURAL NETWORKS

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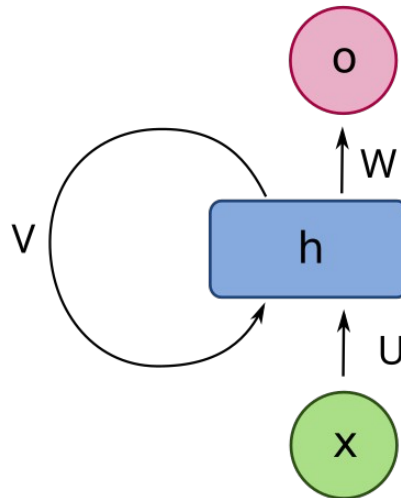
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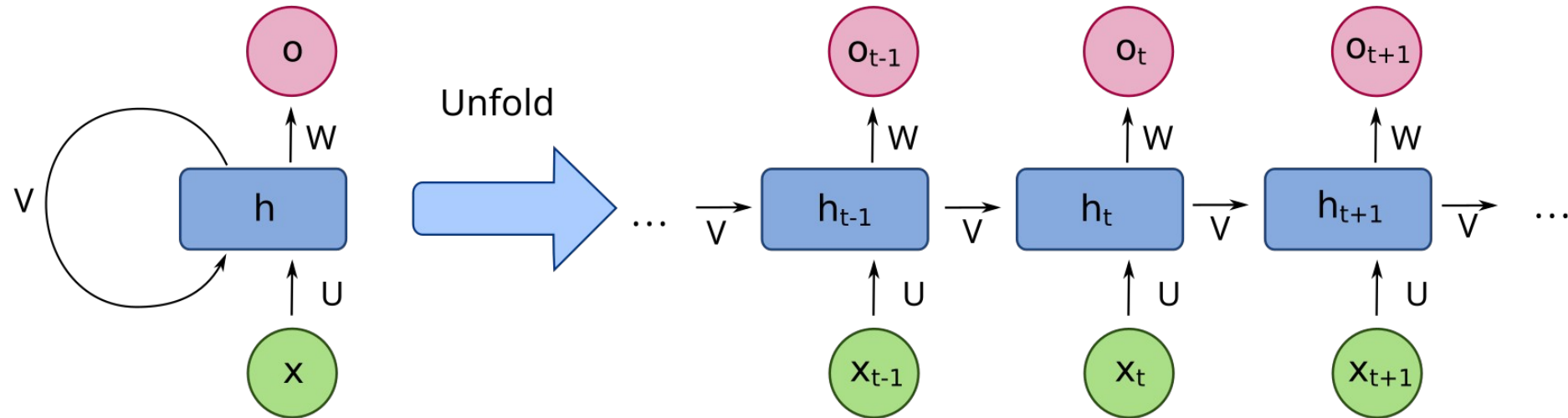
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Recurrent Neural Nets

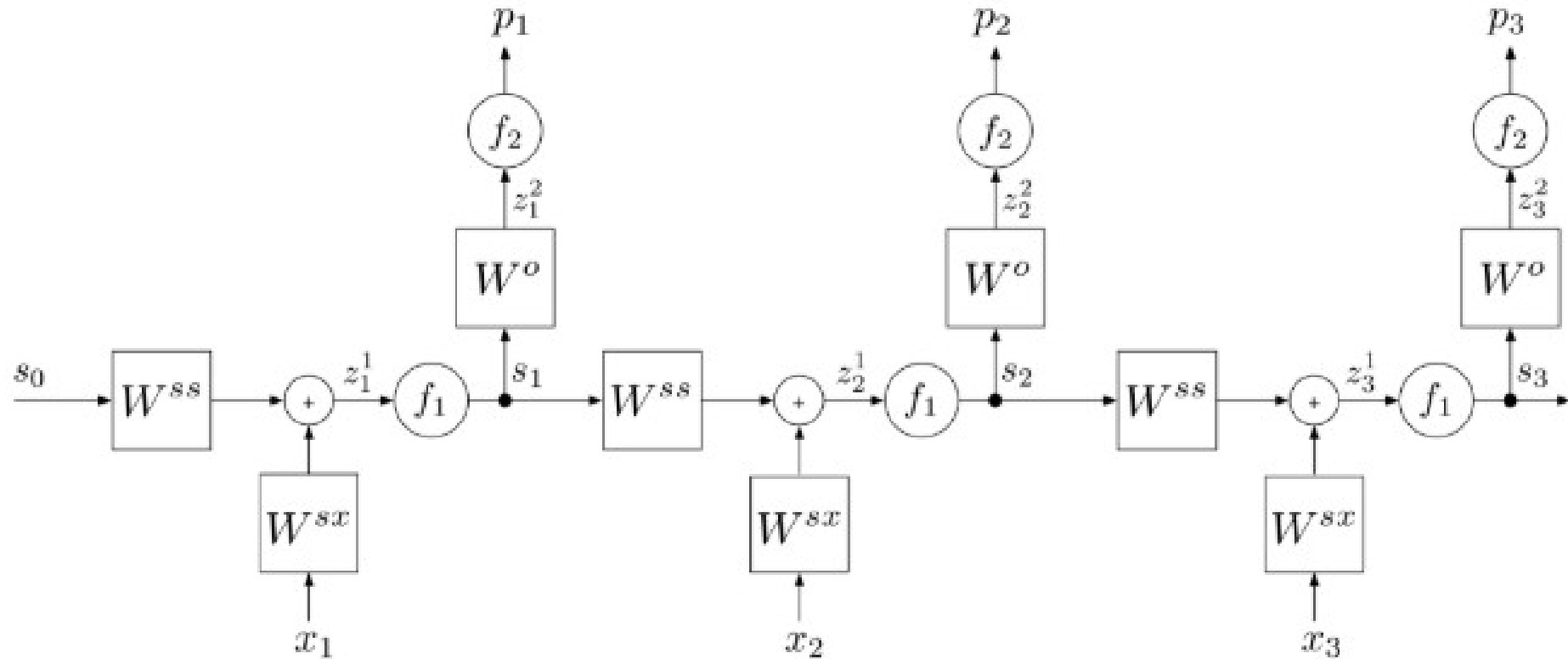
- Idea: Capture effect time has on data by combining new input with current state.
- How it's implemented: "looped" neural net which receives data from input at a given time and the current state



Recurrent Neural Nets



Recurrent Neural Nets



Recurrent Neural Net: Formalization and Loss

- Let's say we want to predict a sequence.

We have pairs of x_t and y_t for a given time step.

So for a given sequence of inputs, we want a certain sequence of outputs

- Loss function is loosely defined – can be anything that measures similarity between predictions and actual, summed across timesteps

RNN: Inductive Bias

- Data has a temporal relationship
- Last state has a large effect on what the next state will be (recency bias)

Chart Review

Encounters Notes **Labs** Imaging Cardiology Microbiology Path Procedures Therapy Other Orders Meds Episodes

Preview Refresh (3:25 PM) Select All Deselect All Review Selected Lab Flowsheet Route Add to Bookmarks Route Lab Requestion

Filters Hide Canceled Blood Bank Pathology Exclude Outside Data Microbiology Exclude Micro/Path Final Results Only

Date/Time	Test	Status
Today		
Today at 12:27	Lead, 24hr Urine	Needs to be Collected
Today at 12:18	CBC with Differential	In process
Today at 12:07	Comp Metabolic Panel	In process
1 Year Ago		
06/11/2019 12:54	Comp Metabolic Panel	Future
06/11/2019 12:54	CBC with Differential	Future
06/07/2019 11:44	Comp Metabolic Panel	Future
06/07/2019 11:44	CBC with Differential	Future
04/03/2019 07:19	Comp Metabolic Panel	Pending
04/03/2019 07:19	CBC with Differential	Pending

Northwestern Medicine

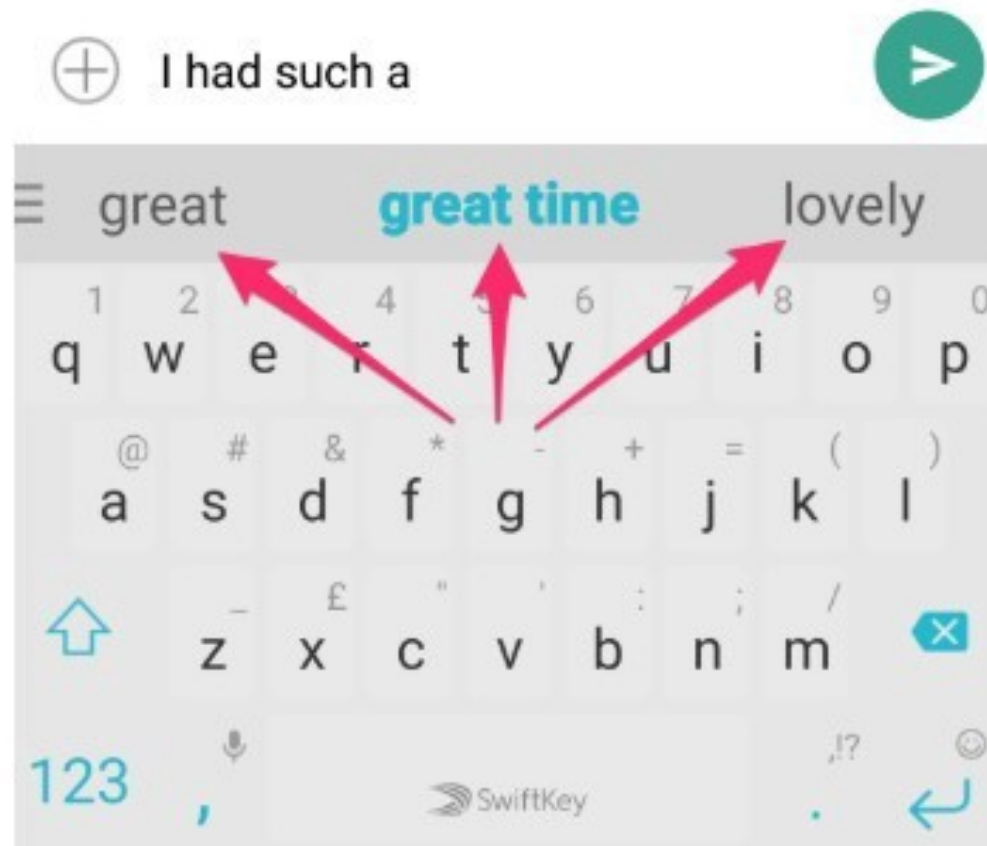
Zztest, Kurt
MRN: 111011393646,
DOB: 8/29/1988,
Sex: M
Visit date: 10/7/2020

NM Family Medicine
10003 US Route 30
Waterman IL 60556-7128

Comp Metabolic Panel
(Accession 20CD-281C00002)
(Order 1163699881)

RNN: Character Prediction

- Let's say we have a sequence of characters and want to predict the next



General Problem Statement

Predict the next word the user wants to type

Technical Problem Statement

Multiclass classification of sequential data – finding the most likely

Data

A ton of user text messages

Loss

Multiclass log loss at each step, summed across network

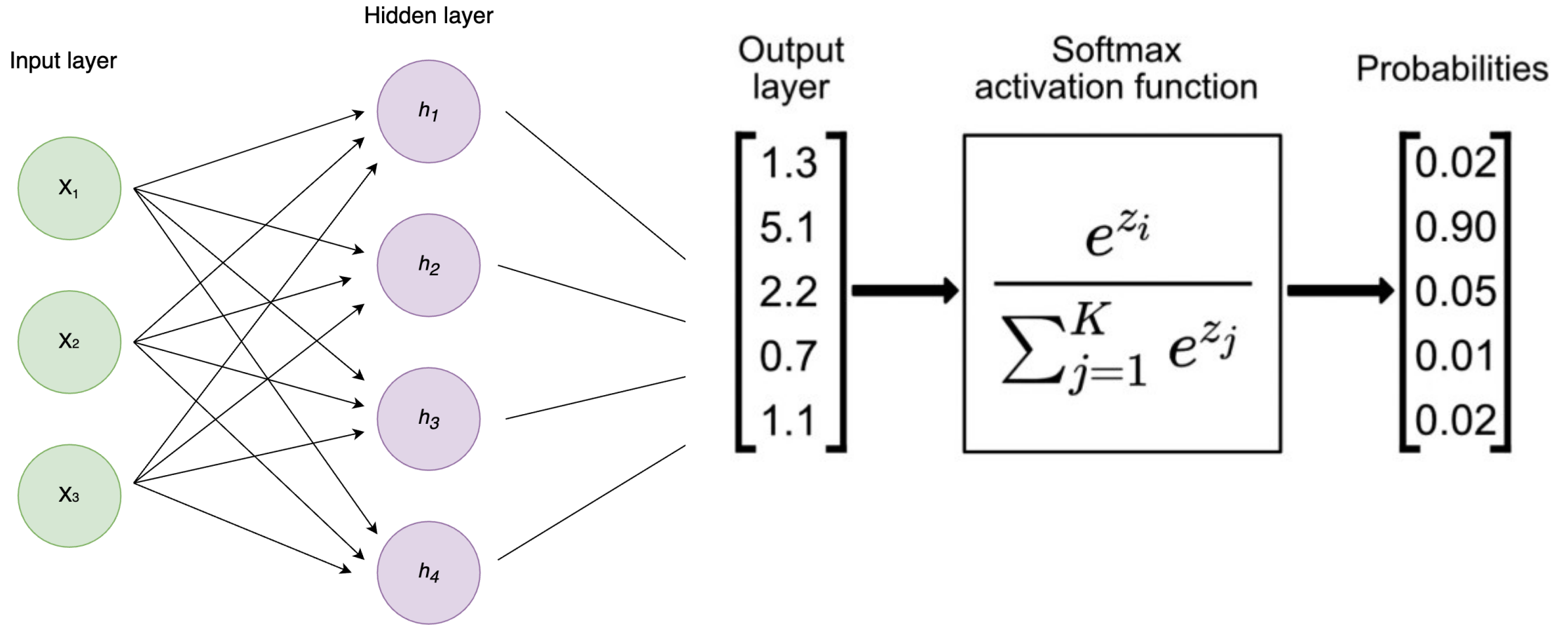
Network

RNN

Training

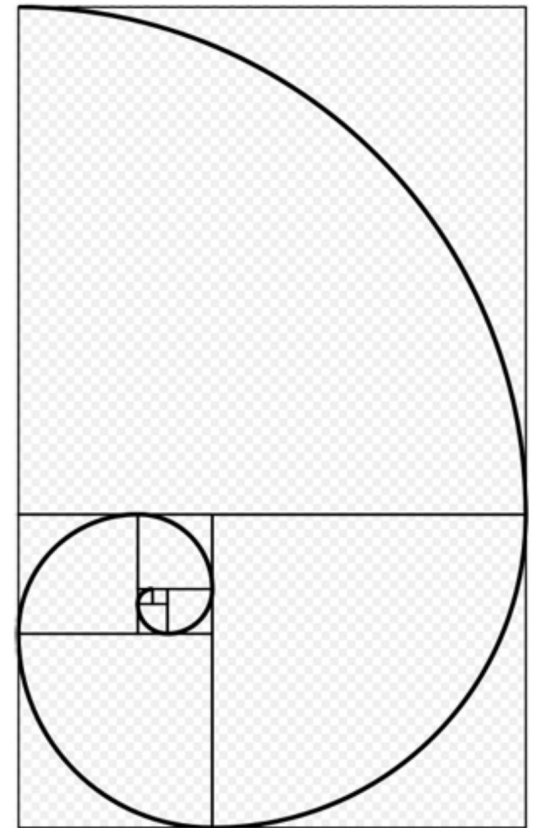
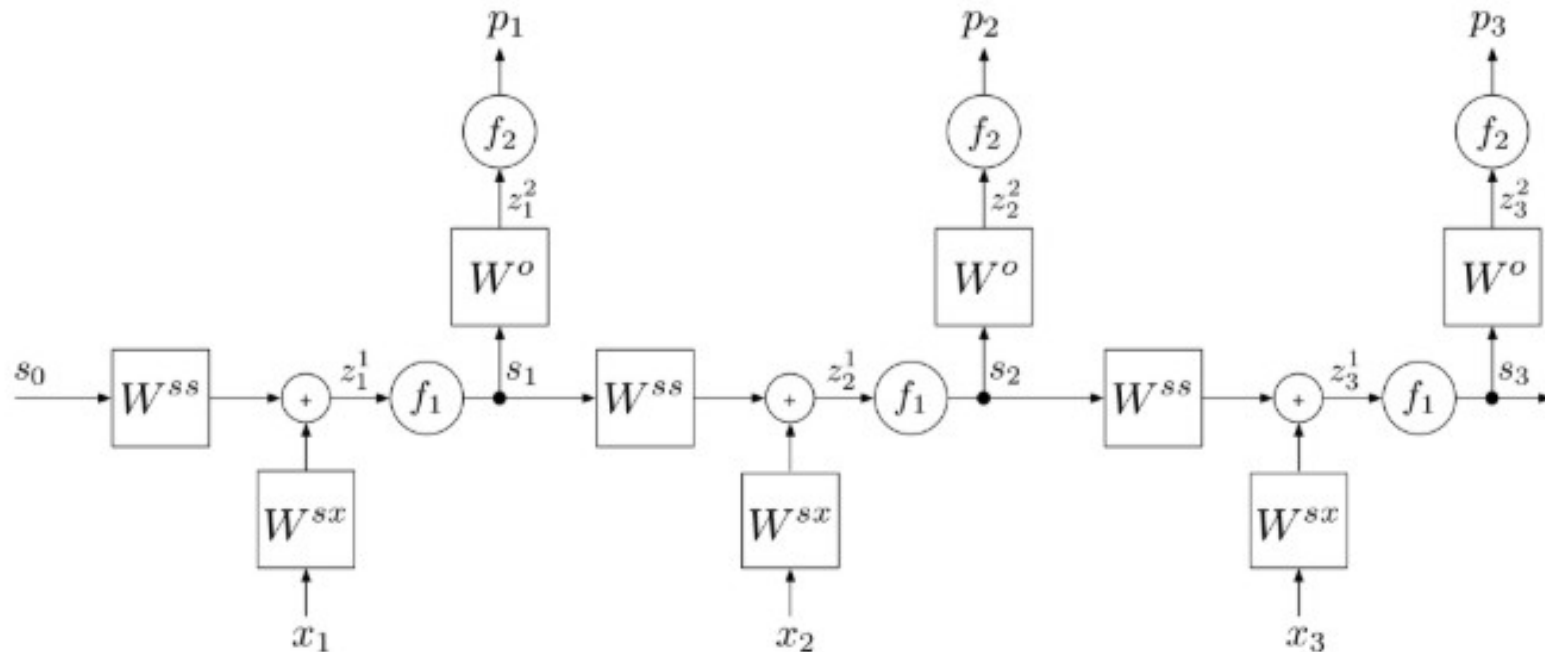
Simple for now

Softmax Function



Issues With RNN

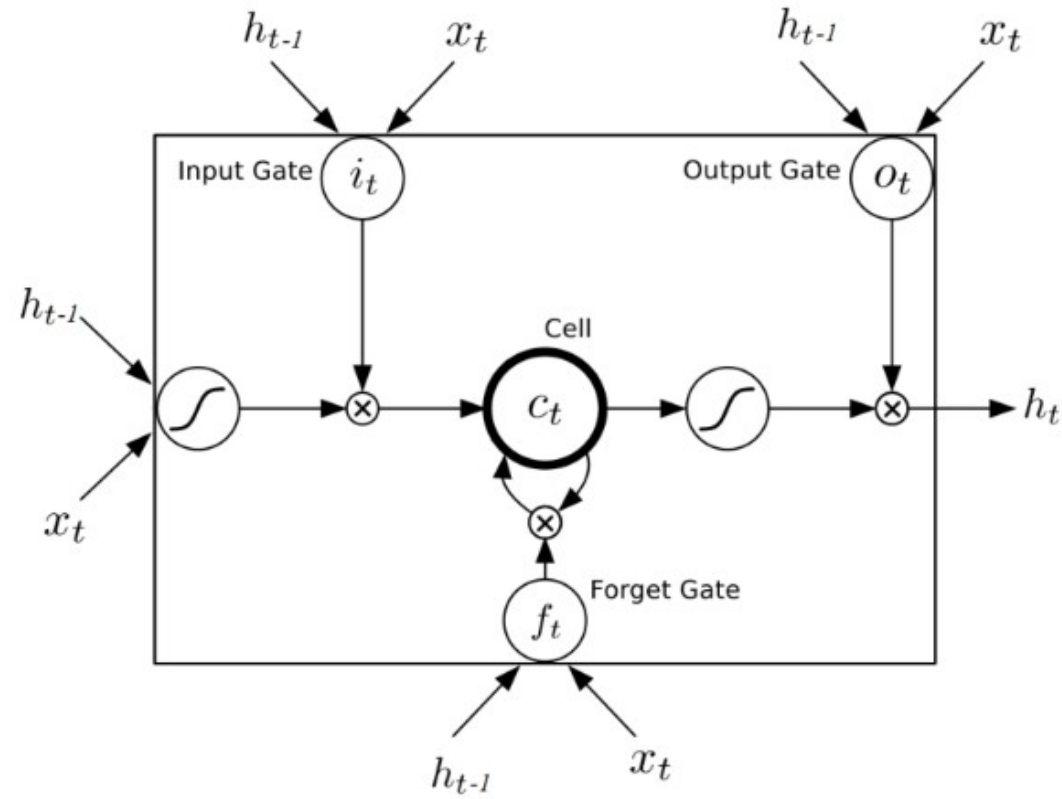
- As we get to a given layer, the input of layer two back means less than the layer one back.



Issues With RNN

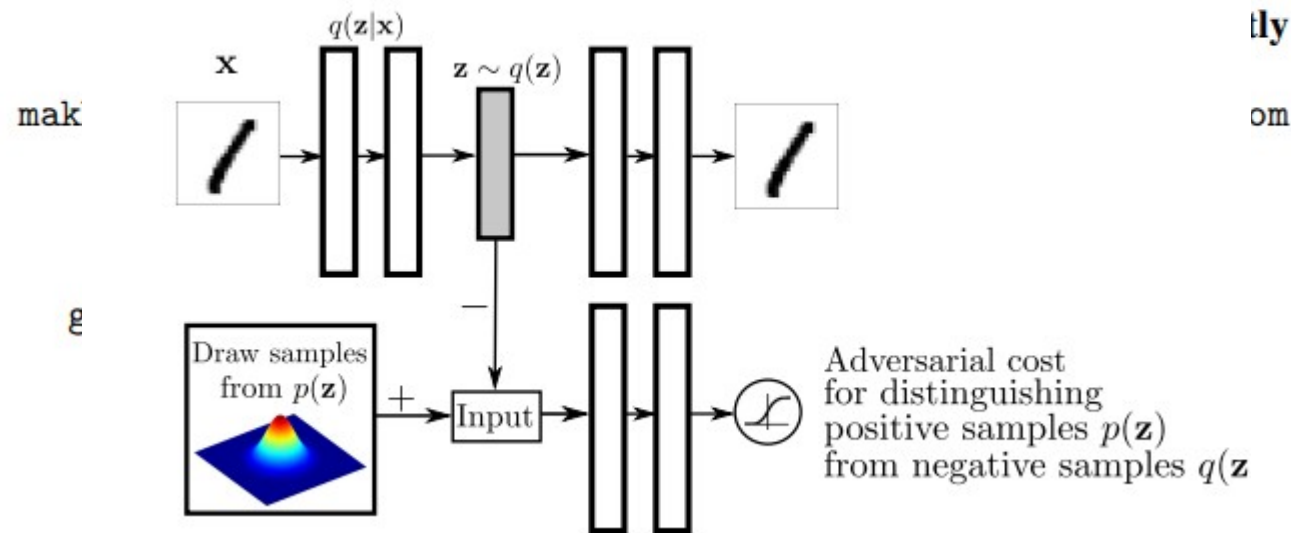
- Variable timestep spacing
- Training must be sequential
- Current state must always have the same matrix size

Long Short Term Memory

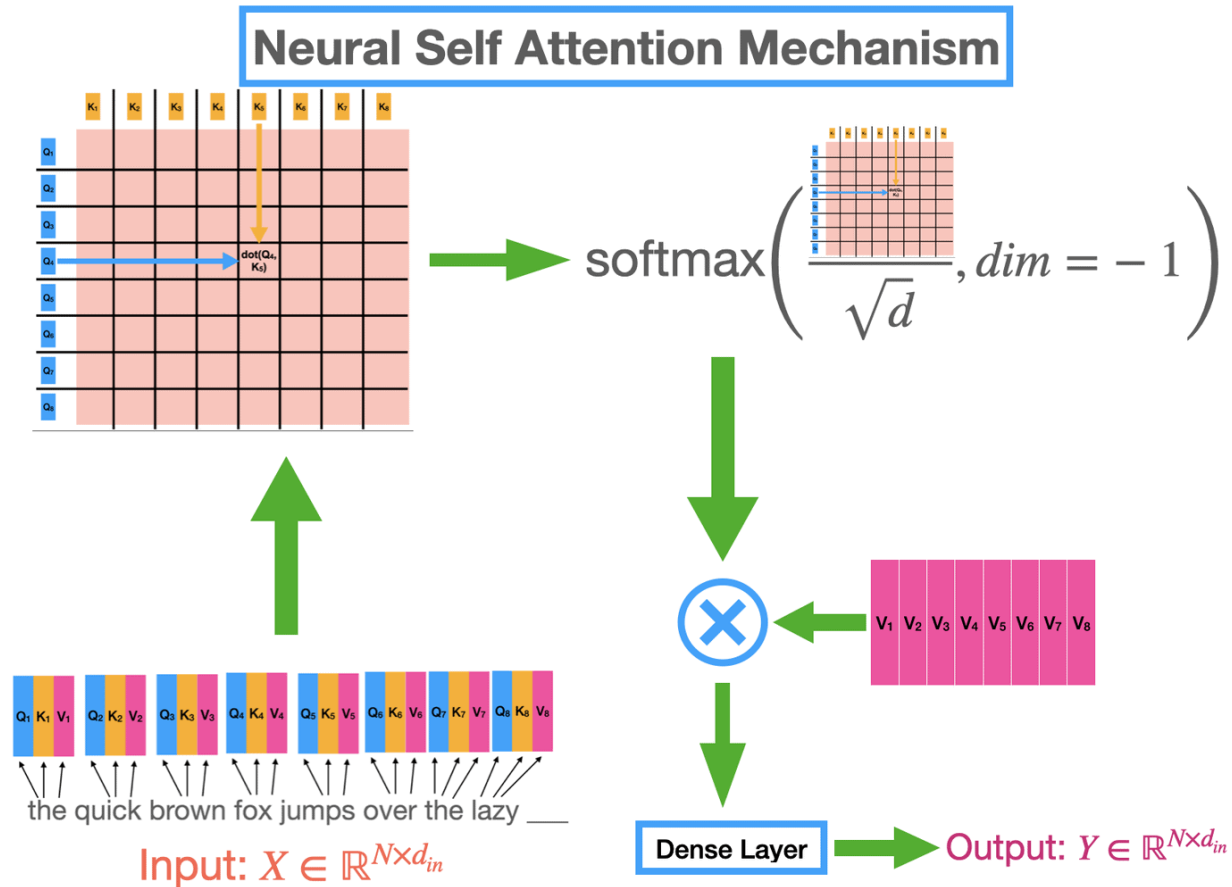


Aside: How to interpret networks in papers

Adversarial Autoencoders



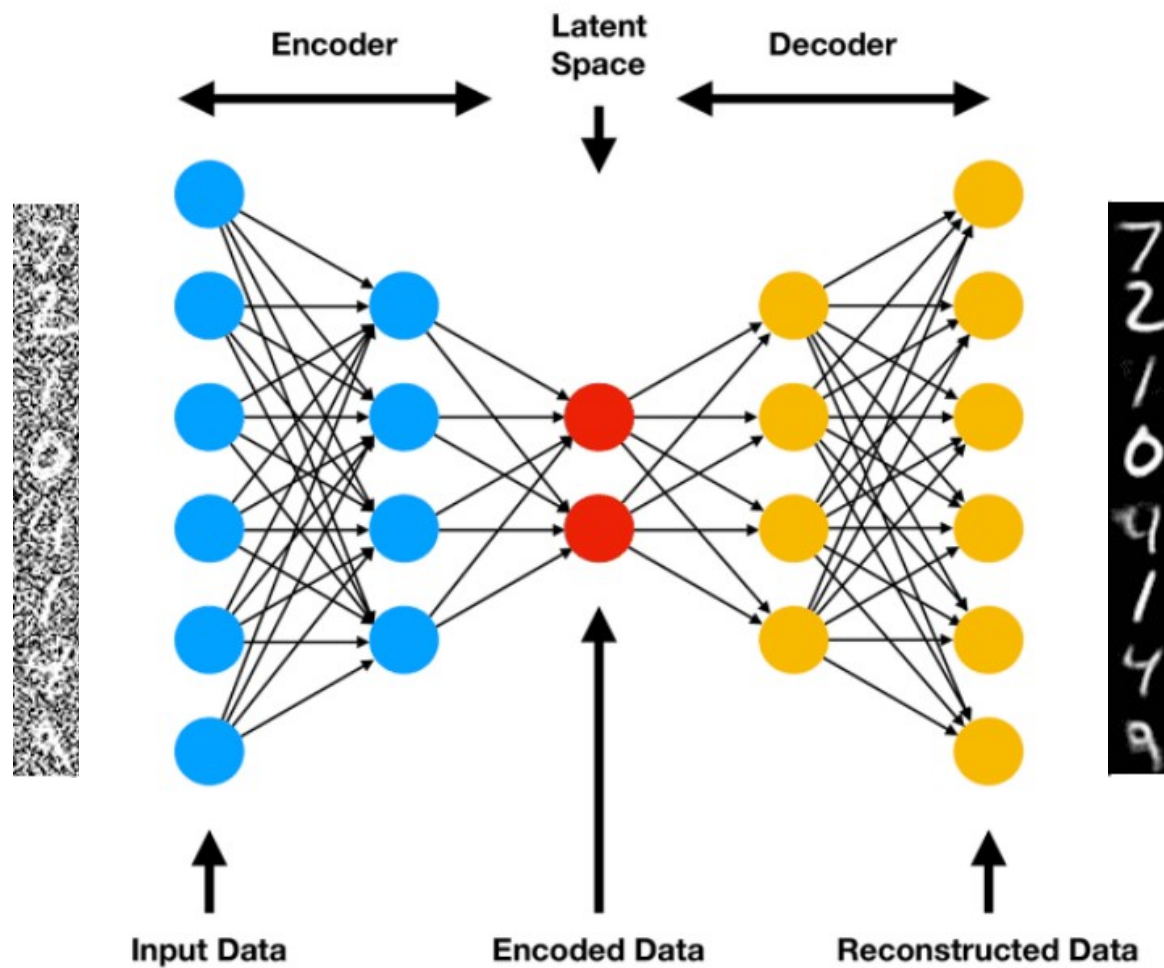
In a few lectures...



Problem Statement

- Learn only the critical information needed to represent data
- Why do this?
 - Remove noise
 - Simplify to human-interpretable patterns
 - Identify critical genes/labs/imaging features that are distinguish example

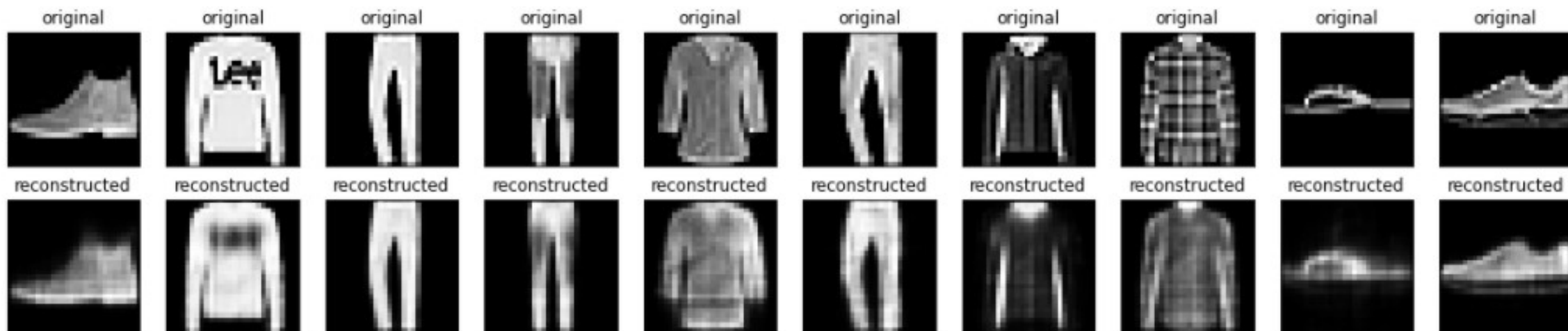
Autoencoders



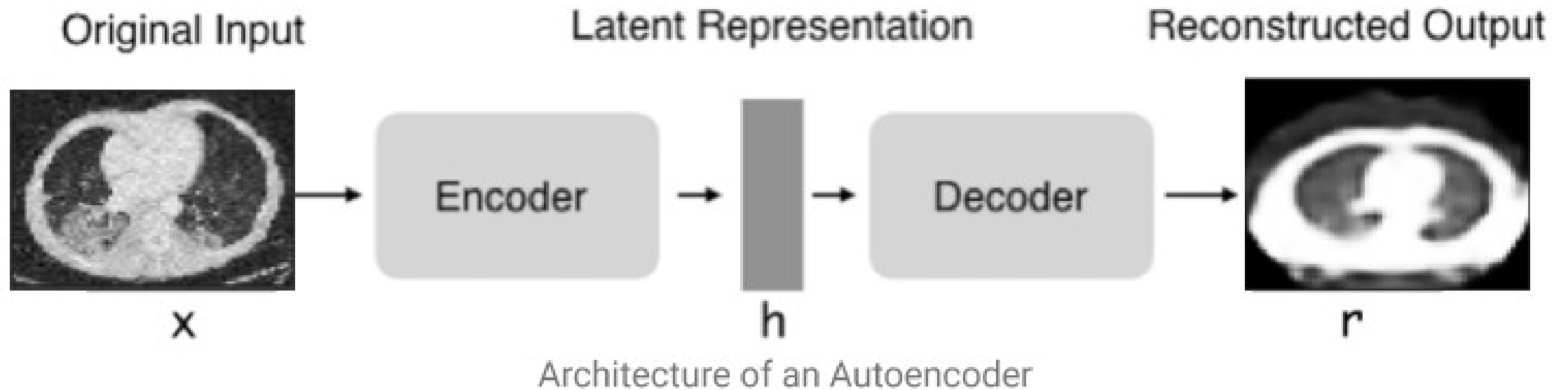
Autoencoder Considerations

- Unsupervised method: Each input is its own expected output
- Tradeoff between bottleneck size and reconstruction fidelity

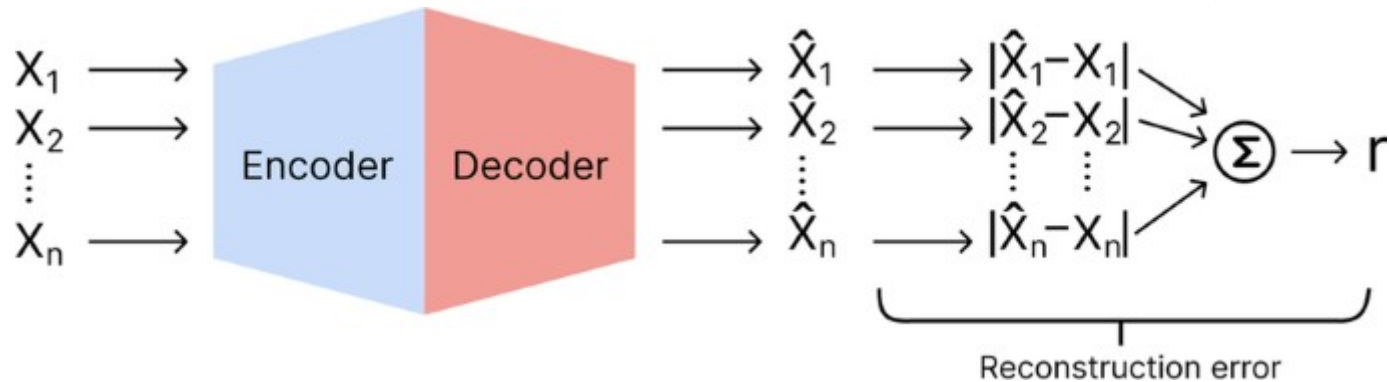
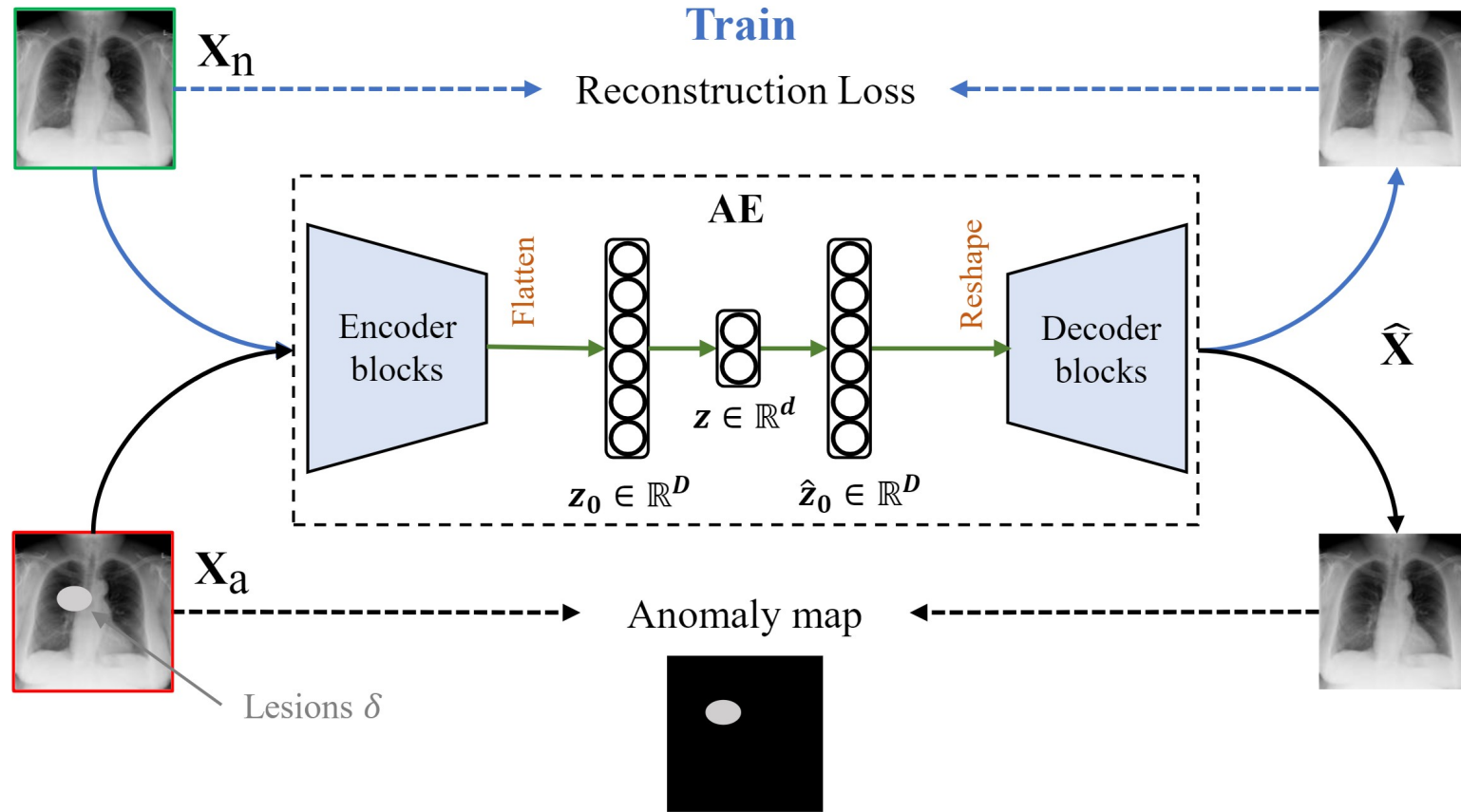
Autoencoder Examples



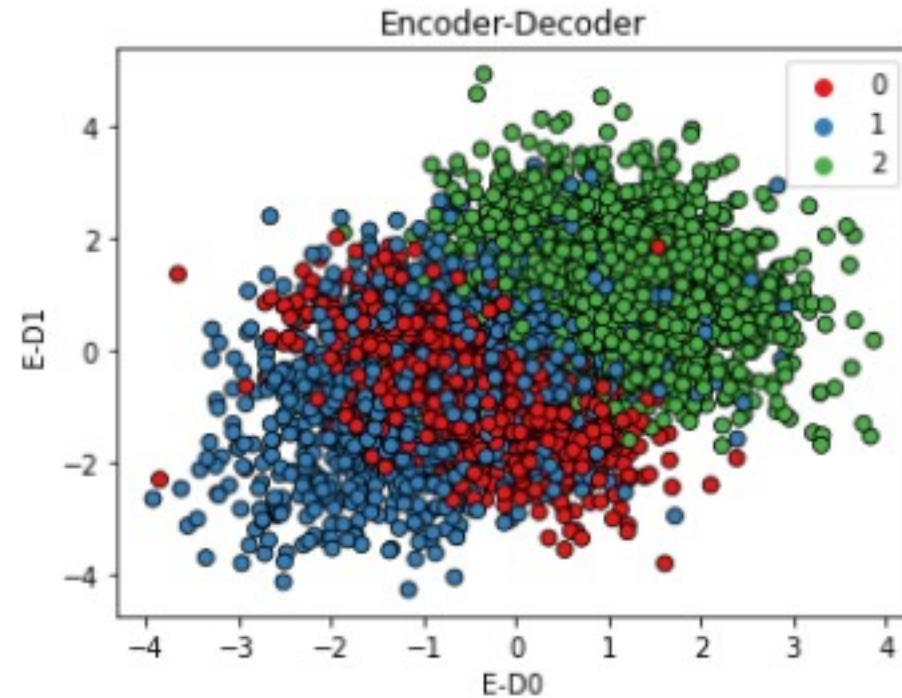
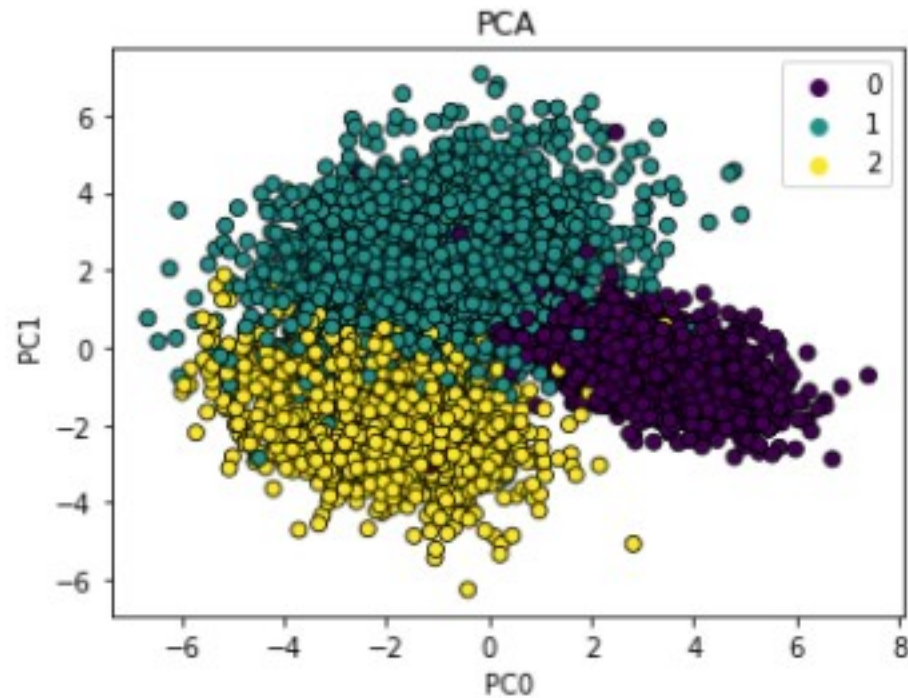
Autoencoder Examples



Autoencoder Loss



Autoencoders vs Principal Component Analysis

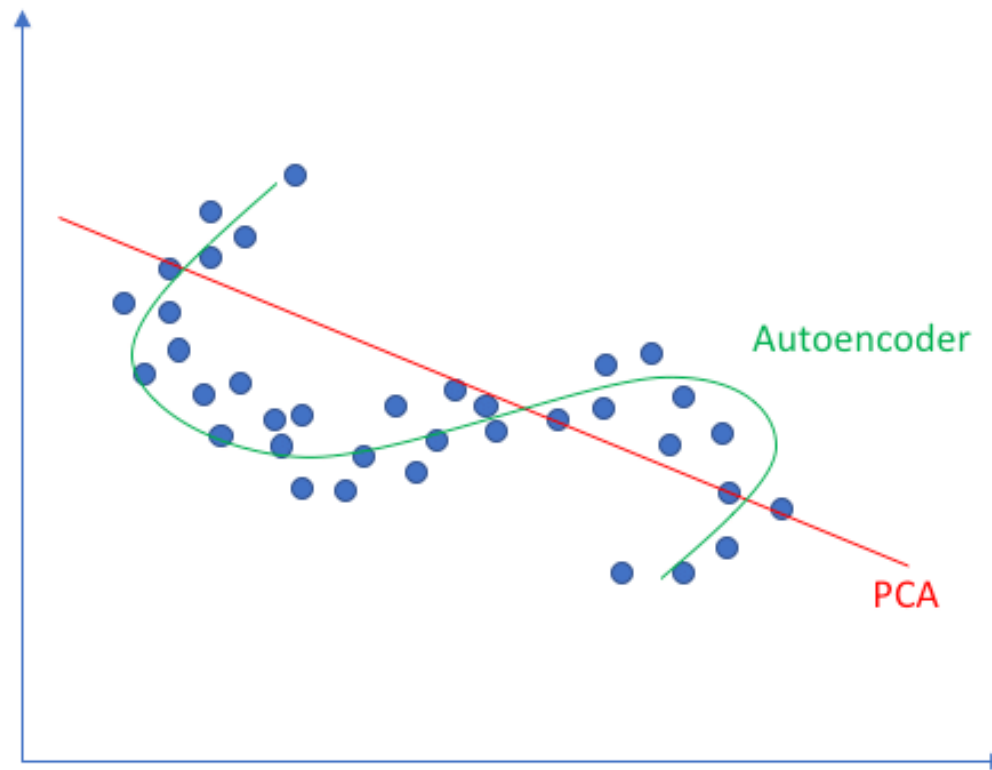




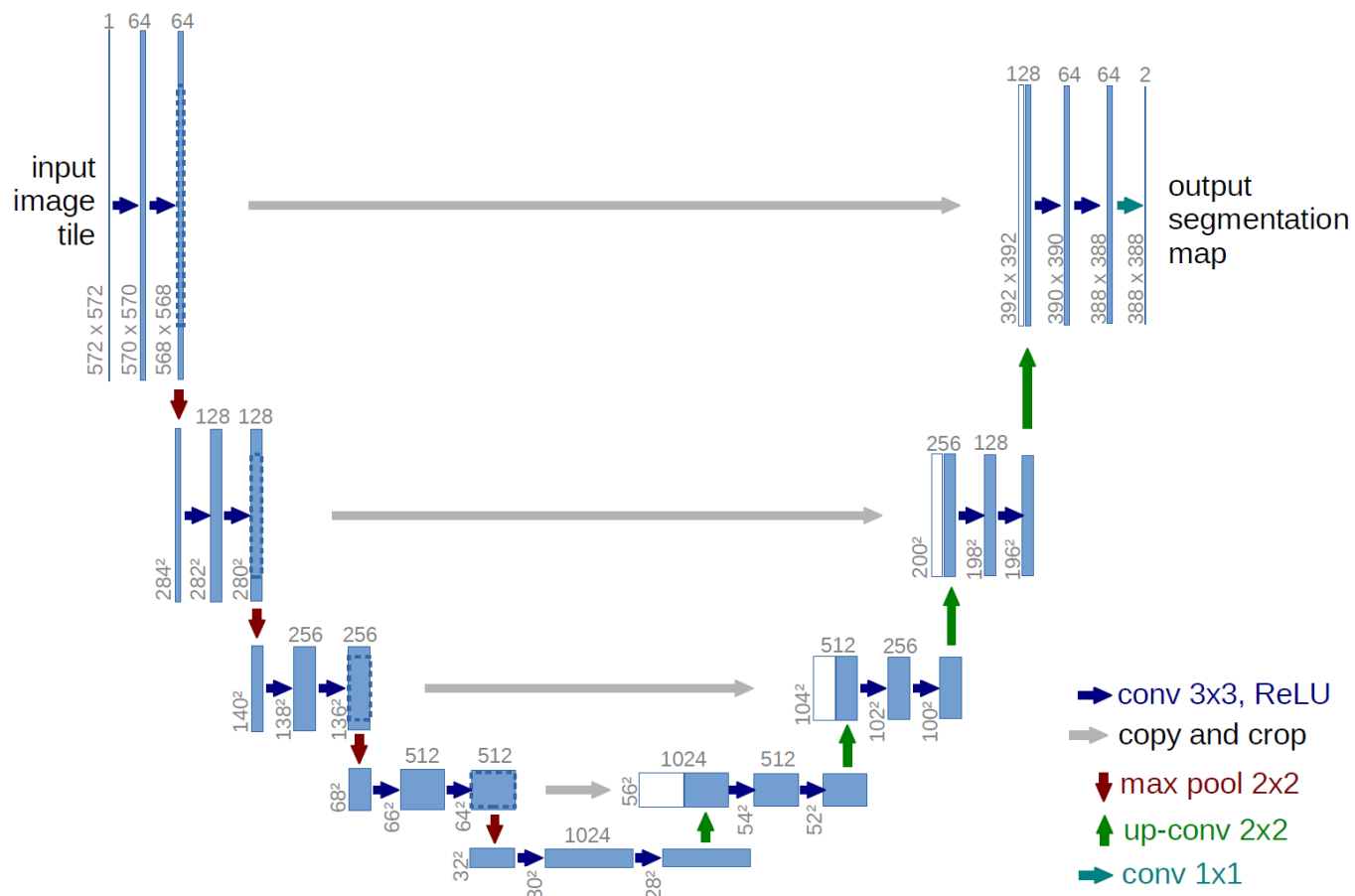
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Autoencoders vs Principal Component Analysis

Linear vs nonlinear dimensionality reduction



Soon...

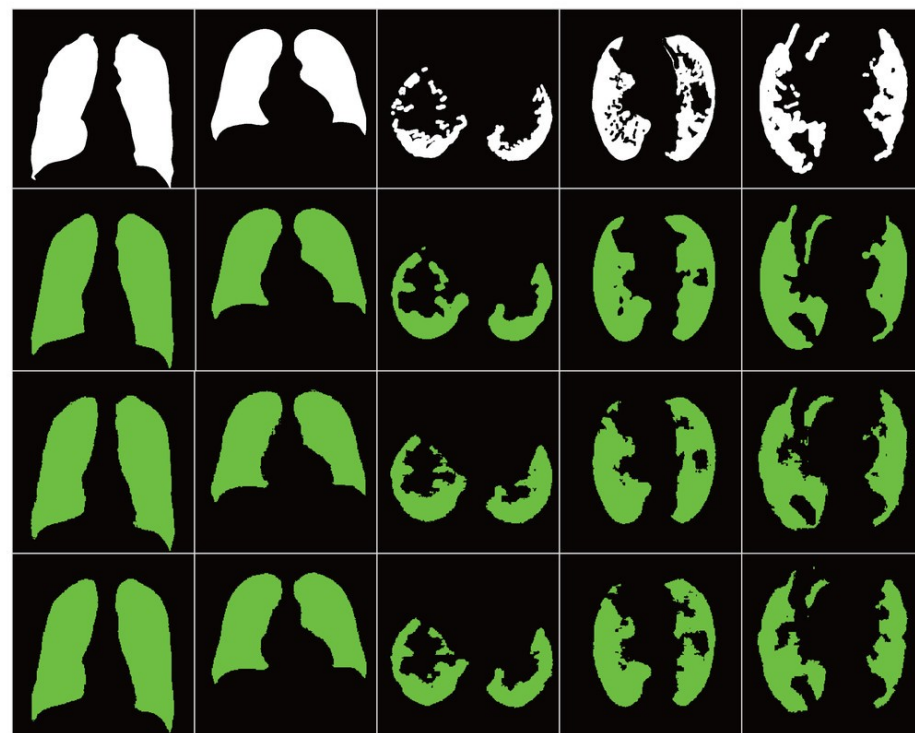


Manual labeling

TransUNet prediction

Swin Unet prediction

Our model prediction



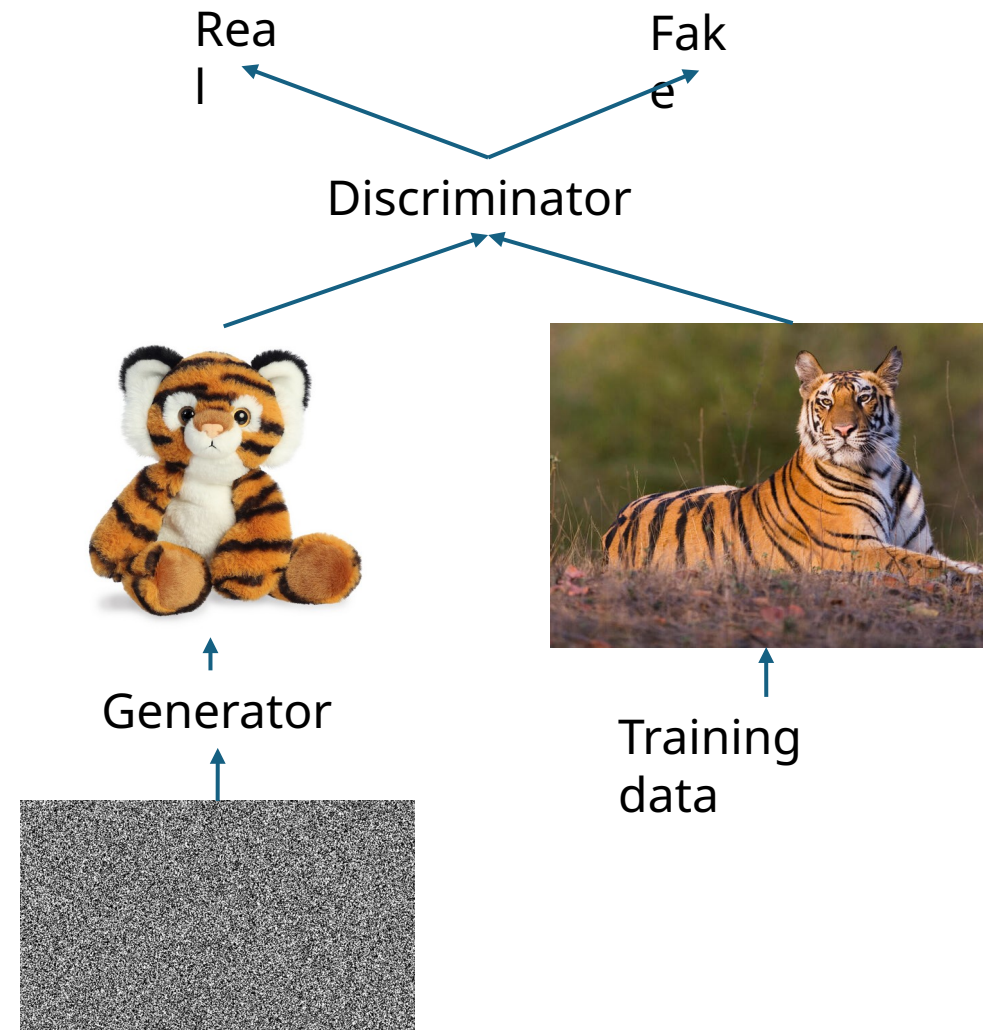
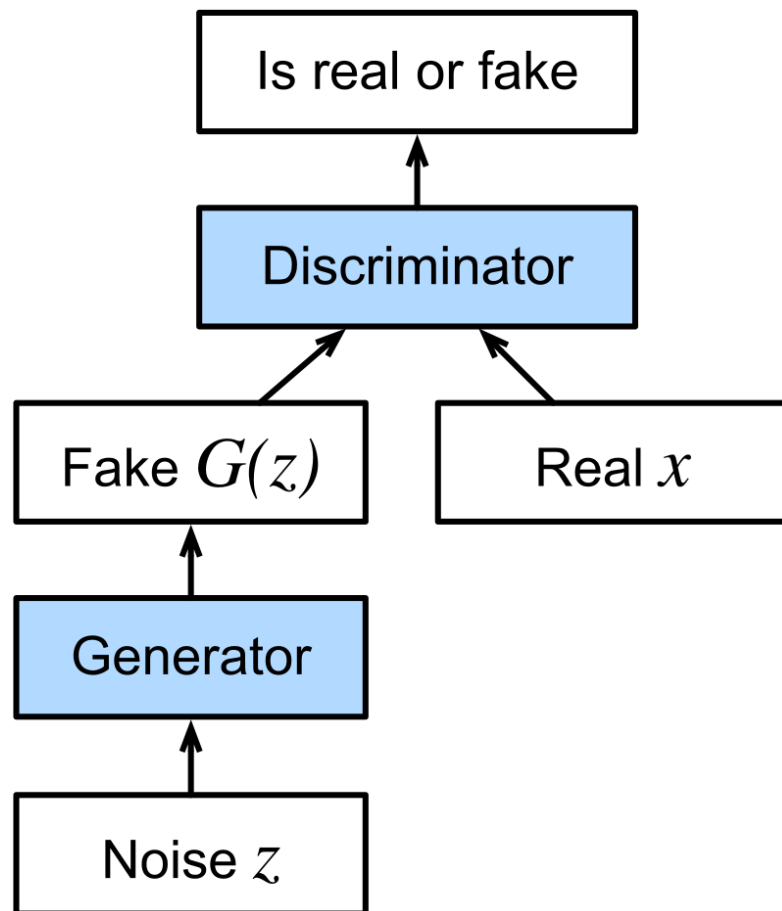
Problem Statement

- We have tons of patient data, but it's not private
- How can we generate data that clinically similar, doesn't lose key info, but is changed in a way that prevents tracing back?

OR

- How do we generate a new antibiotic that is structurally different from an existing one, but works the same?

GAN



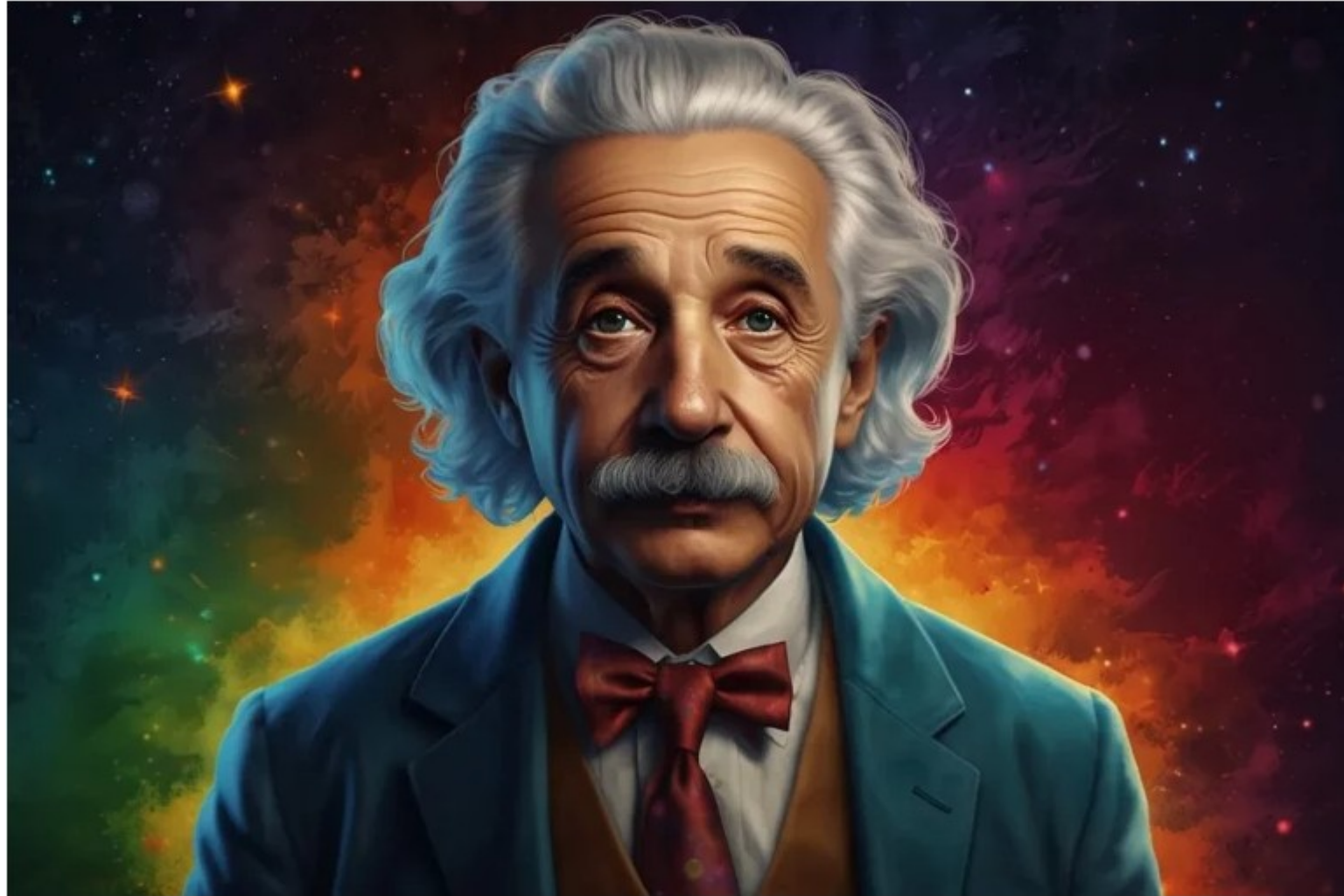
GANs

General Problem Statement	
Generate new data that's indistinguishable from other data	
Technical Problem Statement	
Adversarial binary classification	
Data	Loss
Any group of similar data, Sample one at a time (stochastic)	Binary log loss for both D wants to classify correctly G wants to fool D
Network	Training
As on last slide	Take turns updating weights for both G and D

Mode Collapse

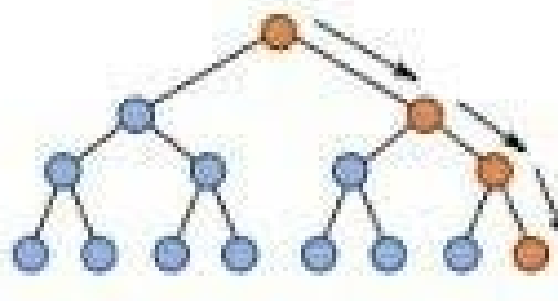
- Error in any generative AI network
- When generated data too highly represents one class
 - Dashes in ChatGPT
 - Always predicting negative on chest x-ray
- Common when AI is trained on AI generated data
- Tradeoff between quality and diversity
 - Regularize!
 - Hyperparameters

AI Generated Images

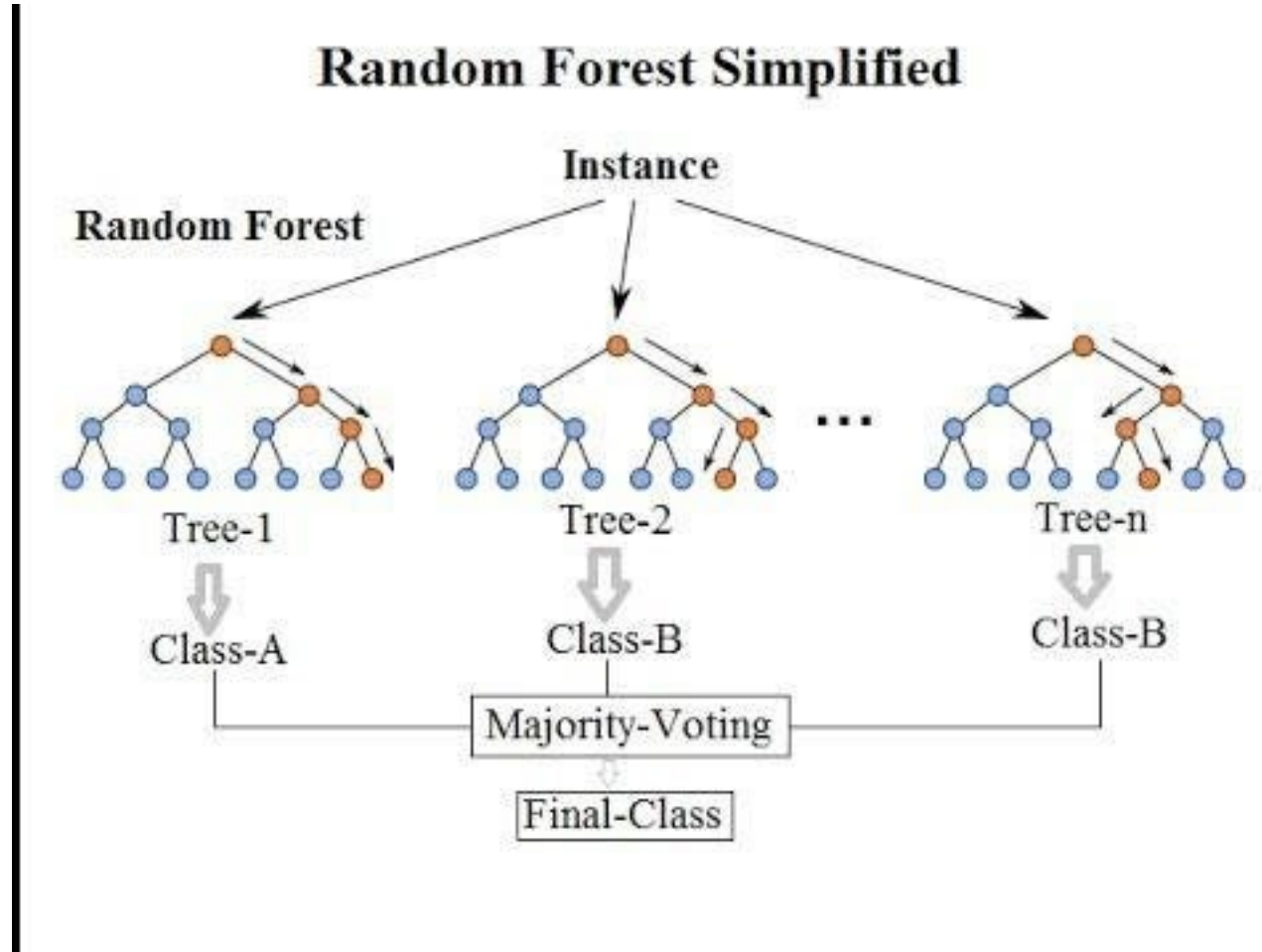


Random Tree

- Not a true neural net model
- Often used in medicine because interpretable
- Plan:
 - Take a bootstrapped data sample
 - Find a matrix operation to split the data in half
 - Repeat until you have the amount of classes you want to predict
 - For a new example, go down the tree and get filtered into a class



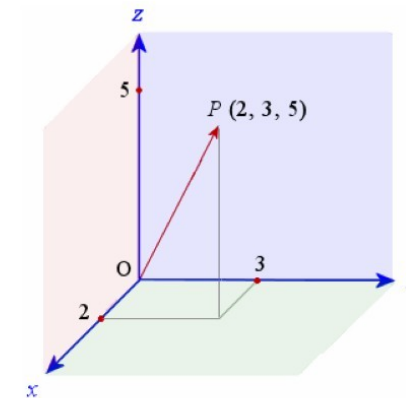
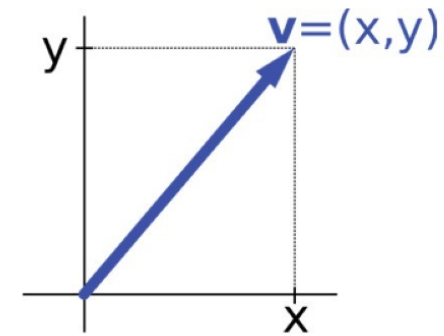
Random Forest



We've come a long way!

Vectors

- Mathematical object with direction + magnitude
 - Direction: like slope
 - Magnitude: length of hypotenuse of triangle w legs x, y
- If no direction, it's
- Can have many di



Thanks!